

Abstract

Transcription factors (TFs) play a crucial role in regulating the transcription of DNA into RNA. TFs bind to regulatory DNA regions, ensuring that specific genes are transcribed at the right time and place. The main targets of TFs are cis-regulatory regions (CRRs), known as promoters and enhancers. Promoters are located around transcription start sites (TSSs), while enhancers can be found far from the genes they regulate. TFs bind promoters and enhancers to recruit other proteins and RNA polymerase II to initiate the transcription. TFs are sequence-specific; their DNA-binding domains (DBDs) interact with short DNA nucleotide sequences called transcription factor binding sites (TFBSs). Understanding gene regulatory mechanisms requires investigating how TFs interact with DNA and identifying the patterns they recognize.

While the binding properties of individual TFs are well-documented, transcription regulation is complex and often involves multiple TFs working together. Different combinations of TFs can interact with the DNA cooperatively by co-binding the same region or forming protein complexes. The distribution of TFBSs within CRRs determines the binding of different TFs and can have a different regulatory effect. The orientation and spacing in between TFBSs is known as DNA motif grammar. Many cellular processes, such as cell differentiation and response to external stimuli, rely on transcription regulation by cooperating TFs. Studying these combinatorial TF bindings is challenging but essential for understanding the full complexity of transcription regulation.

Experimental techniques like chromatin immunoprecipitation followed by sequencing (ChIP-seq) can determine the genomic sequences bound by a TF of interest *in vivo*. Computational methods then model TF-DNA interactions. The most widely used computational model for TF-DNA interactions is the position weight matrix (PWM). PWMs summarize experimentally derived sequences bound by a TF and define a DNA motif recognized by a particular TF. Motifs can be discovered *de novo* from a set of genomic sequences by identifying overrepresented patterns. Computational models of TF-DNA interactions are stored in databases and can be used to predict TFBSs, annotate unknown motifs, or predict TF co-binding events. JASPAR is an example of a database that contains collections of TF binding motifs for TFs from different taxonomic groups.

The main goal of this thesis was to employ computational approaches to explore TF-DNA interactions by (1) discovering new TF binding motifs and updating the JASPAR database while also developing and updating robust pipelines for automated data processing and post-processing, and (2) identifying and characterizing co-binding DNA patterns.

The first objective was achieved by contributing to the two latest releases of the JASPAR database. For both releases, a vast amount of publicly available genomic data was collected to discover DNA binding motifs. These motifs were then manually curated and added to the CORE and UNVALIDATED collections, depending on the available orthogonal support. In addition to adding new information to the database, existing profiles were revised to ensure metadata quality. The final sets of motifs were used for downstream analyses, with results available on the JASPAR website. Additionally, the database update relied on complex software components, which were extensively revised and updated for each release.

The second objective was achieved by developing a pipeline called COBIND to discover TF co-binding patterns. The discovery method is based on non-negative matrix factorization (NMF) and automatically identifies potential TF co-binding patterns relative to known TFBSs. The pipeline was applied to 5,699 TFBS datasets for 401 TFs across seven species. The discovered co-binding patterns were shared among TFs from the same structural family and are likely functionally relevant, as they exhibit higher evolutionary conservation compared to isolated TFBSs. Furthermore, open chromatin data and single-molecule footprinting data supported the co-binding predictions in corresponding human cell lines and mouse embryonic stem cells, respectively.